Yeritmary Rodríguez Delgado

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
M.S. in Automated Science: Biological Experimentation. — *Expected May 2027*
University of Puerto Rico at Cayey — Cayey, PR
B.S. in Biology. — *June 2025*

SKILLS

Programming & Backend: Python, FastAPI, SQL, SQLite, NumPy, pandas, PyTorch, scikit-learn, Git, REST APIs
Computer Vision & Image Processing: OpenCV, scikit-image, U-Net, YOLOv8, CellPose, image segmentation, object detection, feature extraction, quantitative image analysis
Clinical Imaging & Visualization: NIfTI, 3D medical image visualization, volumetric imaging, image segmentation, anatomical labeling, centerline extraction, morphological analysis, coordinate-aware image analysis
Frontend & Visualization: React, TypeScript, JavaScript, Vite, VTK, interactive dashboards, 3D visualization
Scientific Imaging: Fluorescence microscopy, confocal microscopy, Imaris, ImageJ/Fiji, image acquisition, segmentation validation, measurement reproducibility

EXPERIENCE

Robotics Engineer — 2026 – Present
Magnify Biosciences — *Pittsburgh, PA*

- Build Python-based robotic workflows and diagnostic tools for automated scientific experiments, translating user requirements into reliable system behavior.
- Diagnose failures across software, sensing, hardware configuration, and physical workflows through structured root-cause analysis.
- Develop an AI-assisted error-tracing system to surface failures, guide troubleshooting, and improve operator visibility into automated workflows.

Post-baccalaureate Research Fellow, Quantitative Imaging — Jul 2024 – Jun 2025
Carnegie Mellon University — *Pittsburgh, PA*

- Built reproducible quantitative imaging workflows for fluorescence and confocal microscopy datasets spanning 21 experimental targets.
- Applied segmentation, localization analysis, feature extraction, intensity quantification, and morphological measurements to convert complex image data into reproducible quantitative outputs.
- Investigated variability across image acquisition, preprocessing, segmentation, and analysis parameters to improve measurement reliability.
- Standardized image-analysis procedures and documented workflows to improve reproducibility across experimental conditions.

PROJECTS

GadgetInspectra – ROS2 Inspection & Operator Interface — 2026

- Built a React, TypeScript, and Vite operator dashboard for a ROS2-based inspection system, visualizing sensor measurements, inspection state, system progress, and automated decisions.
- Designed structured system-state and inspection-result workflows to support reproducibility, debugging, and clear operator feedback.

NeuroVasc Workbench – 3D Clinical Imaging Platform — 2026

- Built a 3D neurovascular imaging platform for visualization and quantitative analysis of Circle of Willis segmentations from medical imaging datasets.
- Implemented volumetric processing, vessel-specific segmentation analysis, centerline extraction, graph-based measurements, and anatomical labeling across 12 cerebral vessel classes.
- Computed quantitative vessel metrics including centerline length, mean and minimum diameter, candidate diameter reduction, surface area, and mesh volume.
- Built a FastAPI backend exposing case metadata, voxel spacing, anatomical labels, and per-vessel measurements through structured API endpoints.
- Developed interactive 3D visualization workflows connecting quantitative measurements with anatomical locations to support interpretation of vessel morphology.

U-Net Image Segmentation – Automation of Scientific Research — Jan 2026 – May 2026

- Developed a U-Net convolutional neural network for segmentation of scientific imaging data using PyTorch.
- Built an end-to-end pipeline for preprocessing, model training, inference, and quantitative evaluation of segmentation predictions.
- Evaluated segmentation performance using quantitative image metrics and analyzed model uncertainty across experimental images.